SpectralNP: A Sensor-Agnostic Foundation Model for Spectral Remote Sensing

Key idea: A single model that accepts spectral measurements from *any* sensor — regardless of number of bands, spectral resolution, or wavelength positions — and naturally produces higher uncertainty when given less spectral information.

Architecture at a glance

```
Input: {(λᵢ, FWHMᵢ, Lᵢ)} — N bands from ANY sensor
    ↓
Band Encoder — learnable wavelength encoding + conditioned radiance
features
    ↓
Spectral Aggregator — Transformer with RoPE on wavelength + Neural
Process dual path
    ↓
z ~ q(z|context) — stochastic latent whose width ∝ 1/N_bands
    ↓
Task Decoders — spectral reconstruction, material ID, atmospheric
estimation
    ↓
Output: predictions + calibrated uncertainty
```

```
✓ All imports loaded
```

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1. Load Trained Model

Load a pre-trained SpectralNP checkpoint. The model has **~5M parameters** and was trained on synthetic spectra (minerals, vegetation, soil, water, man-made materials) passed through atmospheric radiative transfer simulation with random sensor augmentation.

```
Model loaded — 1,598,398 parameters
  d_model=128, layers=4, z_dim=64
  Trained for 40 epochs, final loss: -0.0988
  Material classes: 300
```

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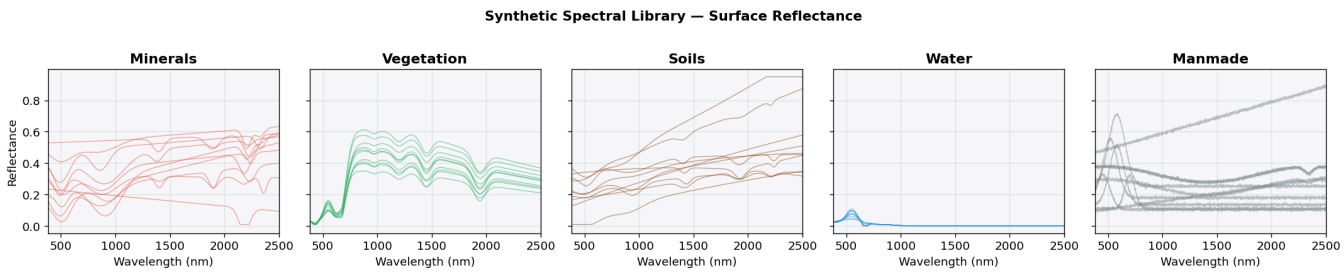
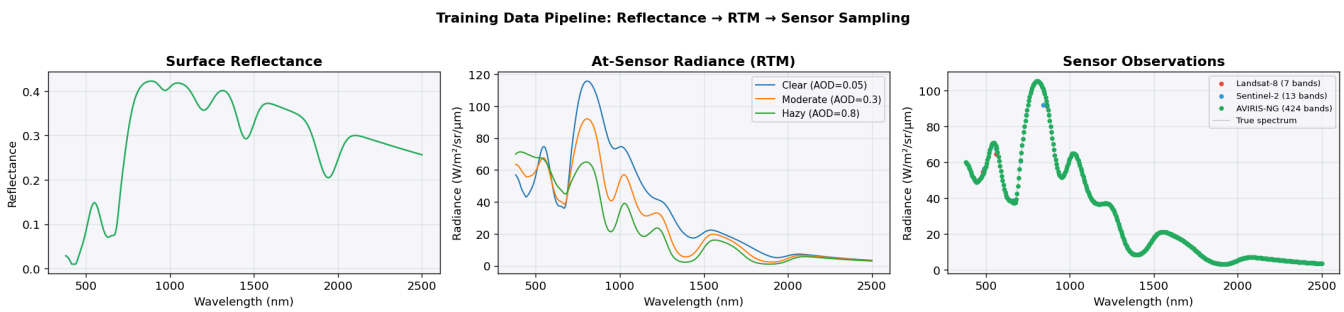
2. Training Data Pipeline

The model is trained on **simulated at-sensor radiance**. We take surface reflectance spectra, pass them through atmospheric radiative transfer, then observe through randomly-generated virtual

sensors.





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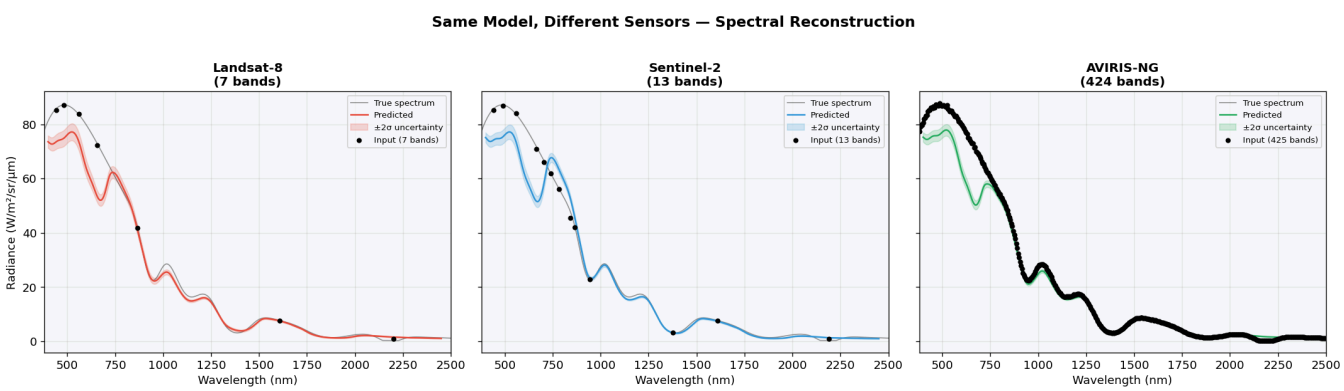
3. Core Capability: Sensor-Agnostic Inference

The **same model** processes data from any sensor. We feed it Landsat-8 (7 bands), Sentinel-2 (13 bands), and AVIRIS-NG (424 bands) — no retraining, no sensor-specific heads.



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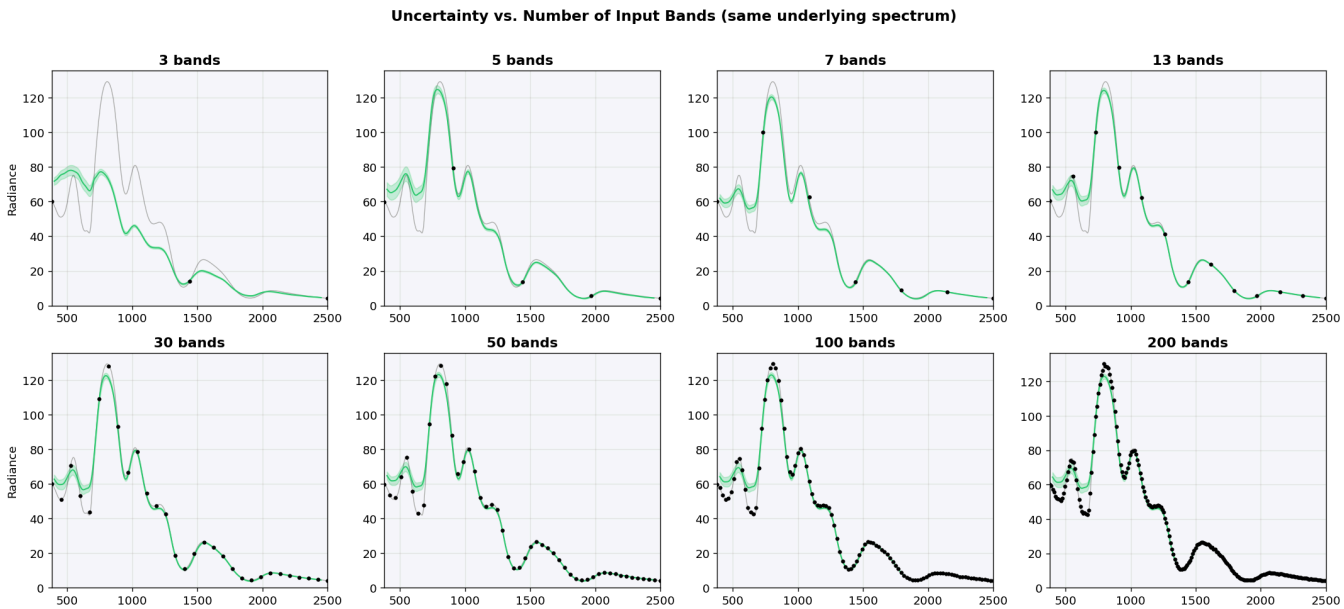
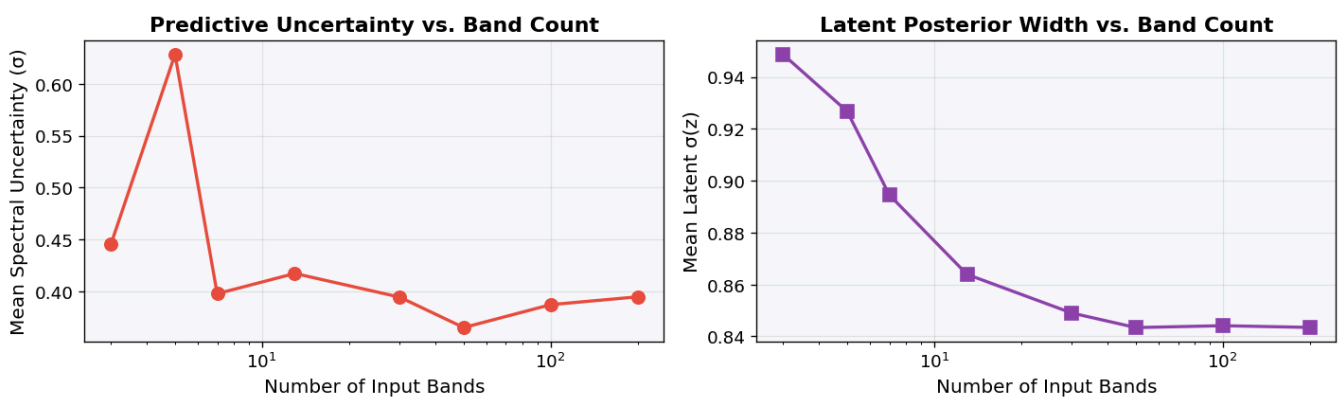
4. Key Property: Uncertainty Scales with Information Content

The **Neural Process latent variable** `z` has a posterior whose width is inversely proportional to the number of observed bands. This is not a bolted-on heuristic — it's a mathematical property of the architecture.

Below we feed the **same spectrum** through virtual sensors with 3, 7, 13, 30, 100, and 200 bands and measure the resulting uncertainty.



More bands → Less uncertainty (by construction)




```
Summary: Uncertainty ratio (3 bands / 200 bands)
  Spectral uncertainty: 1.1x
  Latent posterior σ:   1.1x
```

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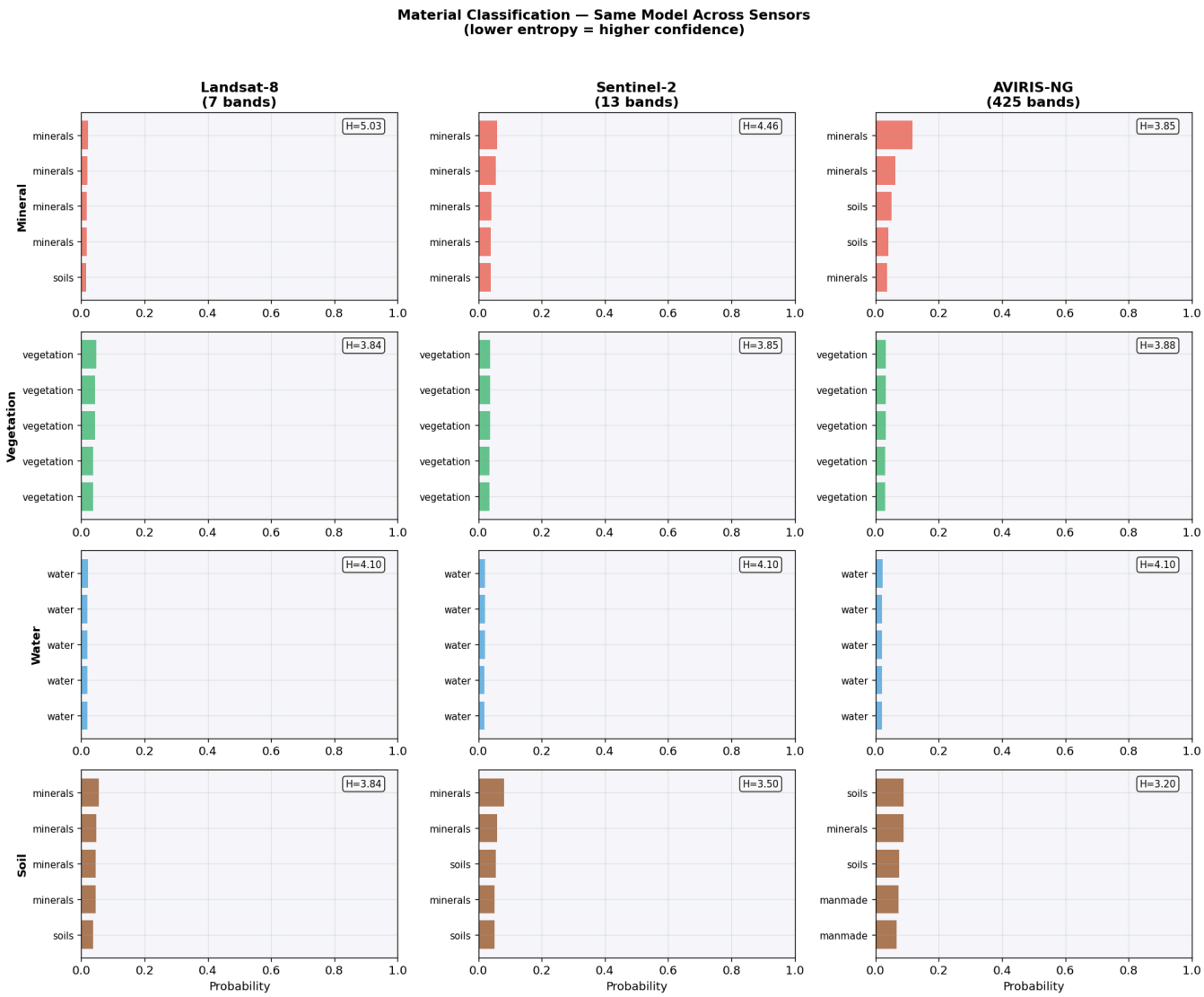
5. Material Identification Across Sensors

The model classifies surface materials from any sensor's input. With more spectral information, the classification entropy decreases (higher confidence).



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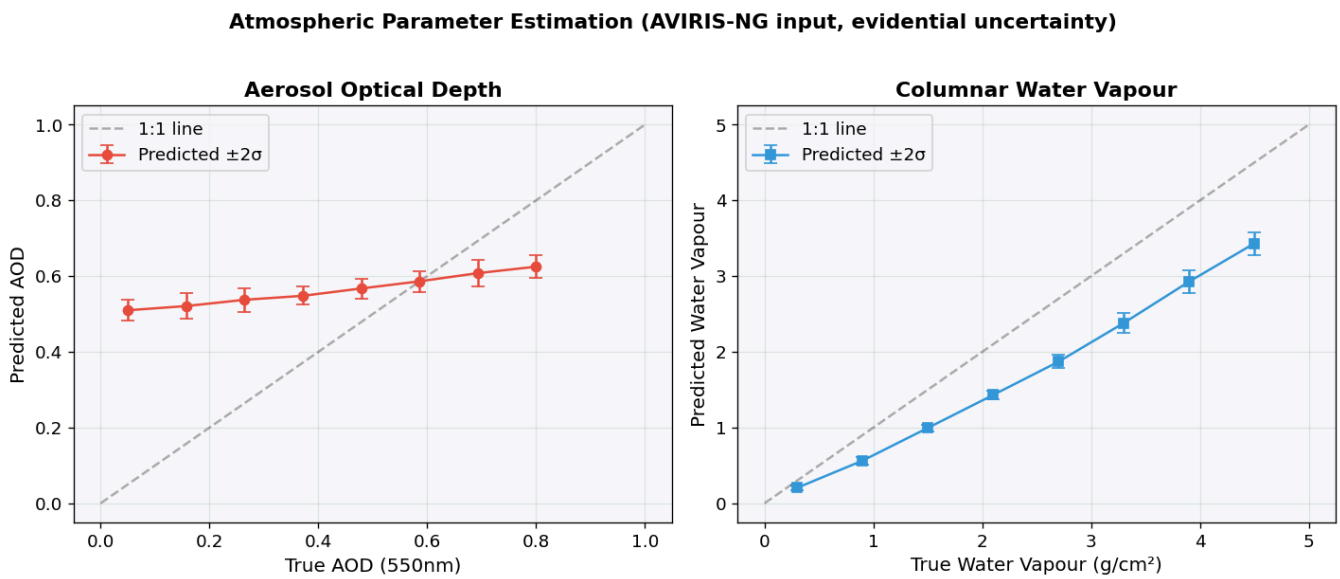
6. Atmospheric Parameter Estimation with Evidential Uncertainty

The atmospheric decoder estimates aerosol optical depth (AOD), water vapour, ozone, and visibility directly from at-sensor radiance — with separate **aleatoric** (data noise) and **epistemic** (model uncertainty) estimates via the Normal-Inverse-Gamma parameterisation.



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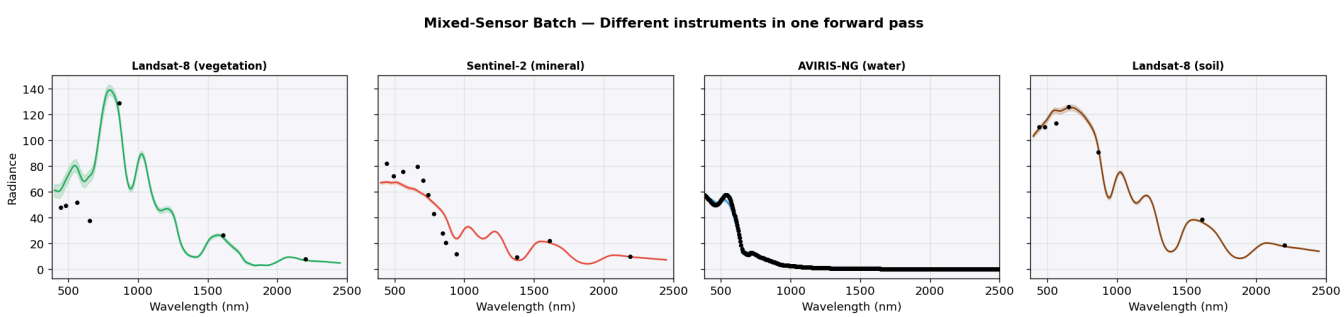
7. Mixed-Sensor Batch Inference

A unique capability: process observations from **different sensors in the same batch**. Each pixel can come from a different instrument.




```
Batch processed: [7, 13, 425, 7] bands per observation (padded to 425)
```

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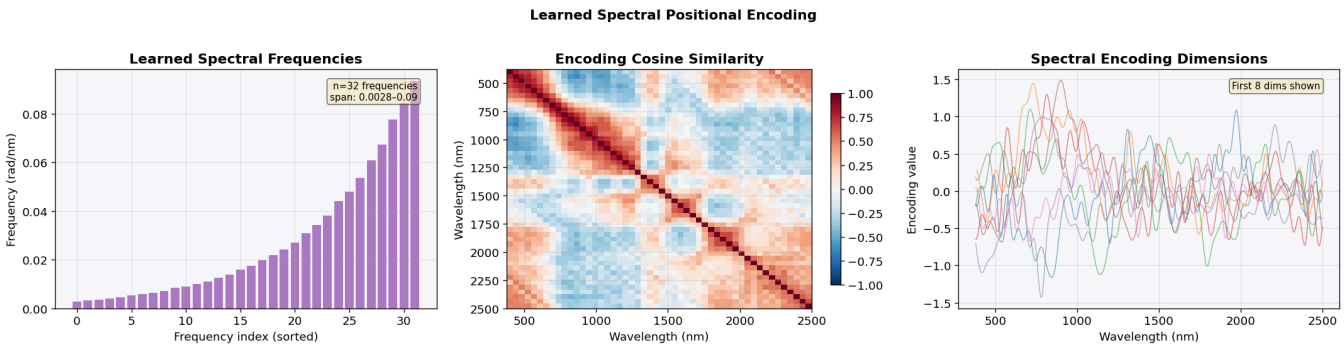
8. Architecture Deep Dive: Why It Works

The architecture combines three ideas that each solve a different part of the problem:



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9. Summary

Feature	How it works
Sensor-agnostic	Each band is a token `(λ, FWHM, L)` — any sensor configuration accepted
Variable band count	Transformer handles variable-length sequences; latent bottleneck gives fixed-size output
Uncertainty ∝ 1/bands	Neural Process posterior widens with fewer context points — by construction
At-sensor radiance	Trained on RTM-simulated TOA radiance; atmospheric effects are learned
Continuous output	DeepONet-style decoder queries any wavelength — not tied to a grid
Multi-task	Shared backbone for spectral reconstruction, material ID, and atmospheric estimation
Evidential uncertainty	NIG output decomposes aleatoric vs epistemic uncertainty

Next steps

- **Scale up**: Train on full USGS Spectral Library v7 (~1300 spectra) with PyARTS full RTM
- **Spatial extension**: Add spatial transformer for image-level predictions
- **Real sensor fine-tuning**: Adapt to Landsat/Sentinel-2/AVIRIS with real ground truth
- **Benchmark**: Compare against sensor-specific models on standard RS benchmarks